M.Tech (CS) Dissertation

Indian Statistical Institute Kolkata

Change-point Analysis of High-dimensional Data Based on Clustering

Banashree Ghosh

CS2138@isical.ac.in

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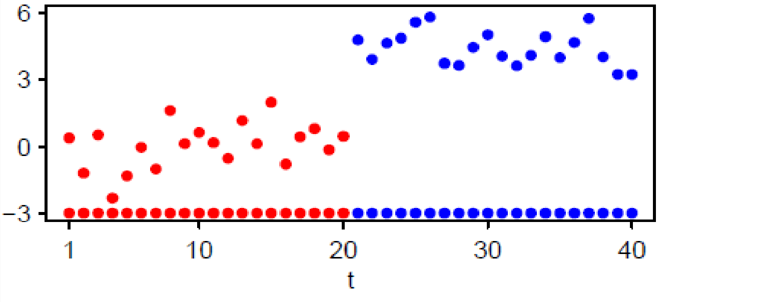
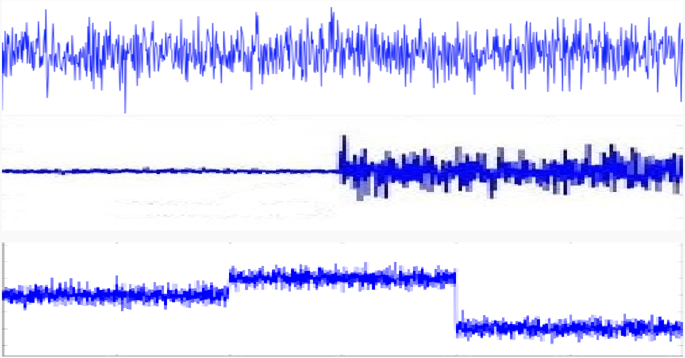
Overview

What is change-point ?

- Change points are the time points when we have an abrupt change in the distribution of the underlying feature variables.
- In a sequence $X_1 \sim F_1, X_2 \sim F_2, \cdots X_n \sim F_n$ of d-dimensional random variables, if $F_{\tau+1}(1 \leq \tau \leq n)$ differs from F_τ, the time-point τ is called a change-point.



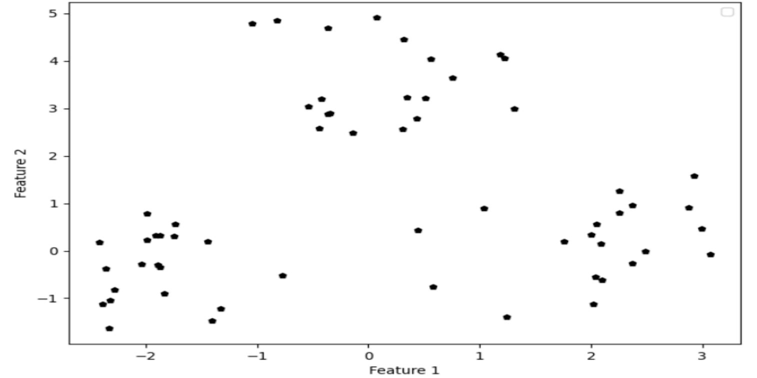
Problem Statement



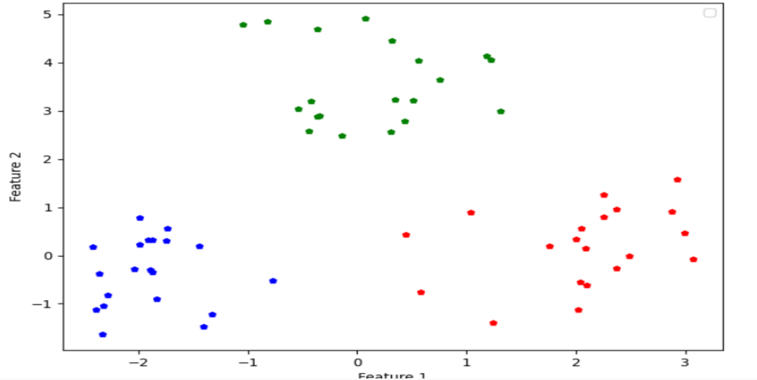
- Aim: Estimation of the number of change-points and their locations.

Detection of change-points using clustering

- Example-1: $X_1, X_2, \cdots, X_{20} \overset{\text{iid}}{\sim} N_2(\mu_1, 0.5\mathbf{I})$,
  $X_{21}, X_{22}, \cdots, X_{40} \overset{\text{iid}}{\sim} N_2(\mu_2, 0.5\mathbf{I})$,
  $X_{41}, X_{42}, \cdots, X_{60} \overset{\text{iid}}{\sim} N_2(\mu_3, 0.5\mathbf{I})$,
  where $\mu_1 = [-2, 0]^T, \mu_2 = [2, 0]^T, \mu_3 = [0, 2\sqrt{3}]$



Detection of change-points using clustering



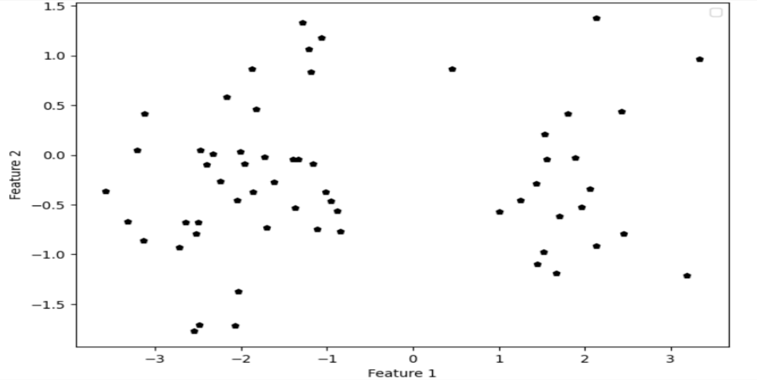
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Implementation of Hierarchical clustering with single linkage

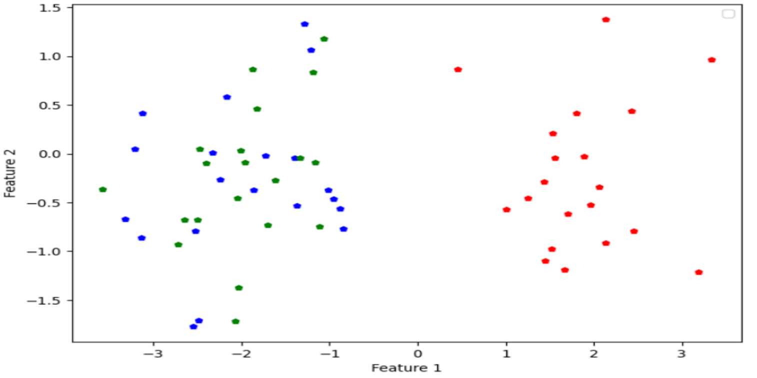
- Pseudo code for hierarchical clustering
  1. Begin with n clusters containing one object, and we will number the clusters 1 through n.
  2. Compute the distance matrix based on the Euclidean distance between two clusters.
  3. Merge the clusters based on the minimum distance between clusters.
  4. Update the distance matrix based on single linkage.
  5. Repeat Step 3-4 a total of n-1 times until only one cluster is left.

Does this method always work?

- Example-2: $X_1, X_2, \cdots, X_{20} \overset{\text{iid}}{\sim} N_2(\mu_1, 0.5\mathbf{I})$,
  $X_{21}, X_{22}, \cdots, X_{40} \overset{\text{iid}}{\sim} N_2(\mu_2, 0.5\mathbf{I})$,
  $X_{41}, X_{42}, \cdots, X_{60} \overset{\text{iid}}{\sim} N_2(\mu_1, 0.5\mathbf{I})$,
  where $\mu_1 = [-2, 0]^T, \mu_2 = [2, 0]^T$



Does this method always work?



1 2 3 4 5 6 7 8 910 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

How to take care of this ?

- Pseudocode for modified Hierarchical clustering algorithm:
  1. Begin with n clusters, each containing one object, and we will number the clusters 1 through n.
  2. Compute the distance matrix based on the Euclidean distance between two consecutive clusters.
  3. Merge the clusters based on the minimum distance between clusters.
  4. Update the distance matrix based on single linkage.
  5. Repeat Step 3-4 a total of n-1 times until only one cluster is left.

Performance of the modified method

- Performance on Example 1 :
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
- Performance on Example 2 :
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
- It performed well in both examples.

Does this method work for high-dimensional data

- Example-3: $X_1, X_2, \cdots, X_{20} \sim N_{100}(\mu_1, 0.5\mathbf{I_{100}})$,
  $X_{21}, X_{22}, \cdots, X_{40} \sim N_{100}(\mu_2, 0.5\mathbf{I_{100}})$,
  $X_{41}, X_{42}, \cdots, X_{60} \sim N_{100}(\mu_3, 0.5\mathbf{I_{100}})$, where
  $\mu_1 = [0, 0, \cdots, 0]^T, \mu_2 = [1, 1, \cdots, 1]^T, \mu_3 = [-1 - 1, \cdots, -1]^T$
- Performance :
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
  30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
  53 54 55 56 57 58 59 60
- Example-4: $X_1, X_2, \cdots, X_{20} \overset{\text{iid}}{\sim} N_{100}(\mu_1, 0.5\mathbf{I_{100}})$,
  $X_{21}, X_{22}, \cdots, X_{40} \overset{\text{iid}}{\sim} N_{100}(\mu_2, 0.5\mathbf{I_{100}})$,
  $X_{41}, X_{42}, \cdots, X_{60} \overset{\text{iid}}{\sim} N_{100}(\mu_1, 0.5\mathbf{I_{100}})$, where
  $\mu_1 = [-1, -1, \cdots, -1]^T, \mu_2 = [1, 1, \cdots, 1]^T$
- Performance :
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29
  30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
  53 54 55 56 57 58 59 60

Does this method work for high-dimensional data?

- Example-5: $X_1, X_2, \cdots, X_{30} \overset{\text{iid}}{\sim} N_{100}(0_{100}, 2\mathbf{I_{100}})$,
  $X_{31}, X_{32}, \cdots, X_{60} \overset{\text{iid}}{\sim} N_{100}(0_{100}, 4\mathbf{I_{100}})$,
- Performance :
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

Reasons for failure

- Let $X_1, X_2, \cdots, X_\tau \overset{\text{iid}}{\sim} N_d(0_d, \sigma_1^2 \mathbf{I_d})$
  $X_{\tau+1}, X_{\tau+2}, \cdots, X_n \overset{\text{iid}}{\sim} N_d(\mu_d, \sigma_2^2 \mathbf{I_d})$,
- By Weak Law of Large Number(WLLN) we can show that as $d \to \infty$,

$$d^{-1} \|X_i - X_j\|^2 \xrightarrow{\text{p}} \begin{cases} 2\sigma_1^2 & \text{for } 1 \leq i, j \leq \tau \\ 2\sigma_2^2 & \text{for } \tau < i, j \leq n \\ \sigma_1^2 + \sigma_2^2 + \mu^2 & \text{for } i \leq \tau < j \text{ or } j \leq \tau < i \end{cases}$$

- Let us define $d_{11} = 2\sigma_1^2$, $d_{22} = 2\sigma_2^2$ and $d_{12} = \sigma_1^2 + \sigma_2^2 + \mu^2$
- Location Problem (Examples 3 and 4): $d_{11} = d_{22} < d_{12}$.
  Distance between two observations from the same distribution is smaller than the distance between two observations from the different distributions.
- Scale Problem (Example 5): $d_{11} < d_{12} < d_{22}$.
  Distance between two observations from the same distribution is larger than the distance between two observations from the different distributions.

What is the remedy ?

- Euclidean distance does not work properly in the data set with different dispersion structures. So, we use a new dissimilarity measure called Mean Absolute Difference of Distances (MADD)
- MADD : For a set consisting of n observations $X_1, X_2, \cdots, X_n$, MADD between two observations X_i and X_j as defined as

$$\delta_0(X_i, X_j) = \frac{1}{n-2} \sum_{k=1, k \neq i,j}^{n} \big| \|X_i - X_k\| - \|X_j - X_k\| \big|$$

High dimensional behavior of MADD

- If $X_1, X_2, \cdots, X_\tau \sim N_d(0_d, \sigma_1^2 I_d)$
  $X_{\tau+1}, X_{\tau+2}, \cdots, X_n \sim N_d(\mu_d, \sigma_2^2 I_d)$, where $\mu_d = (\mu, \mu, \cdots, \mu)^T$,
  By Weak Law of Large Number(WLLN), we can show that as $d \to \infty$,
  $$d^{-1/2}(n-2)\delta_0(X_i, X_j) \xrightarrow{\text{p}} \begin{cases} 0 & \text{for } 1 \le i, j \le \tau \\ 0 & \text{for } \tau < i, j \le n \\ d_{12}^\star & \text{for } i \le \tau < j \text{ or } j \le \tau < I, \end{cases}$$
  where $d_{12}^\star = (\tau - 1)\left|\sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_1\sqrt{2}\right|$
  $$+(n-\tau-1)\left|\sqrt{\mu^2 + \sigma_1^2 + \sigma_2^2} - \sigma_2\sqrt{2}\right|$$
- If $\mu^2 > 0$ or $\sigma_1^2 \neq \sigma_2^2$, we have $d_{12}^\star > 0$.
  In both location and scale problems, the MADD between two observations from the same distribution is smaller than the MADD between two observations from different distributions.
- Use of MADD led to successful detection of change-points in Examples 3-5.

What if $\mu^2 = 0$ and $\sigma_1^2 = \sigma_2^2$?

- Example-6: $X_1, X_2, \cdots, X_{30} \overset{\text{iid}}{\sim} t_4(0_{100}, \mathbf{I_{100}})$,
  $X_{31}, X_{32}, \cdots, X_{60} \overset{\text{iid}}{\sim} N_{100}(0_{100}, 2\mathbf{I_{100}})$,
- Performance:
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

How to overcome this limitation?

- Use MADD based on ℓ_1 distance.
- For a set consisting of n observations $X_1, X_2, \cdots, X_n$, the ℓ_1-distance based MADD between X_i and X_j is defined as

$$\delta_1(X_i, X_j) = \frac{1}{n-2} \sum_{k=1, k \neq i,j}^{n} \left| \sum_{l=1}^{d} |X_i^{(l)} - X_k^{(l)}| - \sum_{l=1}^{d} |X_j^{(l)} - X_k^{(l)}| \right|$$

- Performance on Example 6:
  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

High dimensional behavior of MADD based on ℓ_1-distance

- $X_1, X_2, \cdots, X_\tau \overset{\text{iid}}{\sim} F$,
  $X_{\tau+1}, X_{\tau+2}, \cdots, X_n \overset{\text{iid}}{\sim} G$
  Under some assumptions we can prove that,
  $$d^{-1}(n-2)\delta_1(X_i, X_j) \xrightarrow{\text{p}} \begin{cases} 0 & \text{for } 1 \le i, j \le \tau \\ 0 & \text{for } \tau < i, j \le n \\ d_{12}^{\star\star} & \text{for } i \le \tau < j \text{ or } j \le \tau < i, \end{cases}$$
  where $d_{12}^{\star\star} \ge 0$ and the equality holds if and only if F and G have identical marginal distributions.

References: (a)Soham Sarkar and Anil Ghosh. (2019), On Perfect Clustering of High Dimension, Low Sample Size Data *IEEE Transactions on Pattern Analysis and Machine Intelligence. PP. 1-1. 10.1109/TPAMI.2019.2912599*.

(b) L. Baringhaus and C. Franz. "On a new multivariate two-sample test". *Journal of Multivariate Analysis 88.1 (2004), pp. 190–206.ISSN: 0047-259X*

Assumptions

- We assume that, $X_1, X_2, \cdots, X_n$ are n independent observations d dimensional random variables coming from k_0 populations $F_1, F_2, \cdots, F_{k_0}$. Let μ_i and $\sum_i$ denote the mean vector and the dispersion matrix of F_i, respectively.We also make the following assumptions.
  (A1) The fourth moments of the measurement variables are uniformly bounded.
  (A2) For two independent observation $U \sim F_i$ and $V \sim F_j$,$\sum_{q \neq q'}$ cov$\{(U^{(q)} - V(q))^2, (U^{(q')} - V^{(q')})^2\} =$ **o**(d^2)
  (A3) There exist finite quantities ν_{ij} and ${\sigma_i}^2$ such that $d^{-1}\|\mu_i - \mu_j\|^2 \to \nu_{ij}$ and $d^{-1}trace(\sum_i) \to {\sigma_i}^2$ as $d \to \infty$

Estimation of the number of change points

- So far, we have assumed that the number of clusters/number of change points is to be known, but it should be estimated from the data.
- We have used DUNN index for this purpose.
  DUNN index : For a fixed k, let $C_1, C_2, \cdots, C_K$ be the clusters estimated by our clustering algorithm.
- Define *within cluster distance* for ith cluster C_i,

$$W_K(i) = \frac{1}{|C_i||C_i - 1|)} \sum_{X_r \in C_i} \sum_{X_s \in C_i, r \neq s} \delta_1(X_r, X_s)$$

and *between cluster distance* between C_i and C_j is defined as,

$$B_K(i,j) = \frac{1}{|C_i||C_j|} \sum_{X_r \in C_i} \sum_{X_s \in C_j} \delta_1(X_r, X_s)$$

where $|C|$ denotes the number of samples in cluster C

- Now define $W(K) = max_{1 \leq i \leq K} W_K(i)$ and $B(K) = min_{1 \leq i < j \leq K, i-j=1} B_K(i,j)$. The ***DUNN index*** is defined by,

$$D(K) = \frac{B(K)}{W(K)}$$

Therefore, the number of clusters can be estimated by $\hat{K_D} = argmax_{2 \leq k \leq K} D(k)$.

References: Dunn, J. C. (1973) A fuzzy relative of the isodata process and its use in detecting compact well-separated clusters. J. Cybern., 3, 32–57.

Performance of our proposed method based on DUNN index

	k = 1	k = 2	k = 3	k = 4	k = 5	$k \geq 6$	Total
Example1	0	0	**100**	0	0	0	100
Example2	1	5	**85**	3	4	2	99
Example3	0	0	**100**	0	0	0	100
Example4	0	0	**100**	0	0	0	100
Example5	0	0	**100**	0	0	0	100
Example6	10	**40**	50	0	0	0	90

Table: Number of clusters estimated by DUNN index in **Example 1-**6

Note: Total represents the number of times change point is found.Bold figures indicate the best result in each example.

Theoretical investigation of DUNN index on high-dimensional data

Theorem

Suppose that we have observations from $K \geq 2$ populations which satisfy $(A1 - A3)$, and also assume that K_{opt} is the optimal number of clusters of the base clustering algorithm. Then as $d \to \infty$,$\hat{K_D} \xrightarrow{p} K$($\hat{K_D}$ is the number of clusters estimated by MADD l_1 version)

References: Soham Sarkar and Anil Ghosh. (2019), On Perfect Clustering of High Dimension, Low Sample Size Data *IEEE Transactions on Pattern Analysis and Machine Intelligence. PP. 1-1*

Limitation of *DUNN* index

- **Example 7 :** Let $X_1, X_2, \cdots, X_{60} \sim N_{100}(0, 0.5I)$.

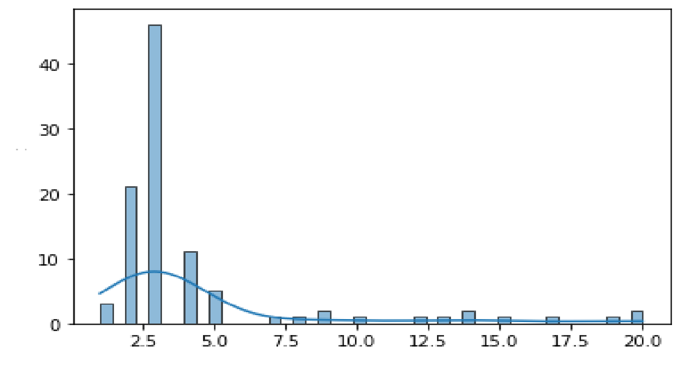


Figure: Example 7

How to overcome this limitation ?

- We define **Penalized *DUNN* index** (PD).
- For any fixed k, it is defined as $PD(k) = \frac{B(k)}{W(k)} - k\zeta(d)$ and $W(k)$ has the same sense as in the *DUNN* index and $B(1) = B(2)$, $\zeta(d) = 0.015 \log d$, is the penalty function.

Contd..

Theorem

Suppose we have observations from $k_{opt} \geq 1$ population(s) which satisfies the assumptions (A1-A3) and the penalized function $\zeta(d)$ is such that $\zeta(d) \to \infty$ and $\frac{\zeta(d)}{d} \to 0$ as $d \to \infty$. If k_{opt} is the optimal number of clusters,then $\hat{K_{PD}} \xrightarrow{p} k_{opt}$.

References: Soham Sarkar and Anil Ghosh. (2019), On Perfect Clustering of High Dimension, Low Sample Size Data *IEEE Transactions on Pattern Analysis and Machine Intelligence. PP. 1-1*

Performance of our proposed method based on Penalized DUNN index

	k = 1	k = 2	k = 3	k = 4	k = 5	$k \geq 6$	Total
Example1	0	0	**100**	0	0	0	100
Example2	0	0	**100**	0	0	0	100
Example3	0	0	100	0	0	0	100
Example4	0	0	**100**	0	0	0	100
Example5	0	0	**100**	0	0	0	100
Example6	10	**50**	40	0	0	0	90
Example7	**100**	0	0	0	0	0	100

Table: Number of clusters estimated by Penalized DUNN index in **Example 1-7**

Note : Total represents the number of times the change point is found. Bold figures indicate the best result in each example

- ***Example 8***, $F_1 = N_{250}(0_{250}, \sum_0)$ *and* $F_2 = N_{250}(1_{250}, \sum_0)$ *have identical scatter matrices* $\sum_0 = ((0.9^{|i-j|}))_{250\times 250}$
  Example 9, $F_1 = N_{250}(0_{250}, \sum_0)$ *and* $F_2 = N_{250}(0_{250}, 3\sum_0)$ *differ in their scatter matrices, where* $\sum_0$ *is same as Example A*
  Example 10 : $F_1 = N_{200}(0_{200}, \sum_1)$ *and* $F_2 = N_{200}(0_{200}, \sum_2)$.
  Here,$\sum_1$, *the first 100 diagonal elements are 1, and the rest are 3 and* $\sum_2$, *the first* 100 *diagonal elements are 3, and the rest are 1*
  Example 11 : $F_1 = N_{200}(0_{200}, 2I)$ and $F_2 = t_4(0_{200}, I)$, the mean vector and the dispersion matrix are the same for both distributions .

Comparing the result with the existing method

Single change point detection

		$\tau = 10$								$\tau = 20$								$\tau = 30$							
	$\lvert\hat{\tau} - \tau\rvert$	0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total
Eample 8	DI	**92**	3	0	0	0	0	0	95	100	0	0	0	0	0	0	100	91	3	0	0	0	0	0	94
	PDI	**88**	0	3	0	0	0	0	91	98	2	0	0	0	0	0	100	83	0	0	0	0	0	7	90
	E-div	**90**	9	0	0	0	0	1	100	86	8	3	0	0	0	3	100	84	6	1	1	0	0	8	100
	Kernel	**94**	6	0	0	0	0	0	100	88	9	3	0	0	0	0	100	91	8	1	0	0	0	0	100
	MST	**68**	10	8	1	4	1	7	99	66	18	6	2	0	1	7	100	64	11	2	6	1	1	11	96
	NNG	**84**	11	2	1	1	0	0	100	83	12	5	0	0	0	0	100	81	12	4	2	1	0	0	100
	MDP	**44**	6	6	6	0	5	30	97	51	13	5	1	2	0	22	94	37	15	5	3	3	1	26	90
Eample 9	DI	**73**	0	0	0	0	5	0	78	78	0	0	0	0	0	0	78	69	0	0	0	0	2	5	76
	PDI	**75**	0	0	0	0	5	0	80	81	0	0	0	0	0	0	81	71	0	0	0	0	2	5	78
	E-div	**10**	4	2	0	1	1	7	25	41	16	7	5	2	0	7	78	27	16	6	3	1	1	3	57
	Kernel	**54**	14	7	3	3	3	16	100	82	10	5	3	0	0	0	100	89	10	1	0	0	0	0	100
	MST	**0**	0	0	0	1	3	28	32	0	0	0	0	0	0	17	17	0	0	0	0	0	0	14	14
	NNG	**0**	0	0	0	0	0	3	3	0	0	0	0	0	0	4	4	0	0	0	0	0	0	3	3
	MDP	**30**	10	7	0	4	10	25	86	36	6	2	1	0	2	21	68	16	7	4	3	2	1	30	63

Table: Frequency distribution of $\lvert\hat{\tau} - \tau\rvert$,where $\hat{\tau}$is the predicted location of change point

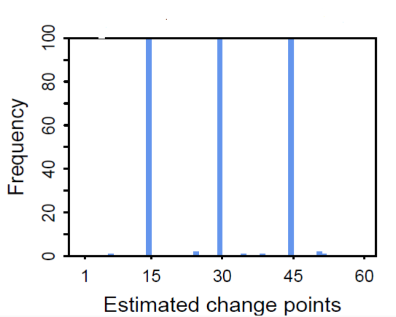
Contd..

	$\|\hat{\tau}-\tau\|$	$\tau=10$								$\tau=20$								$\tau=30$							
		0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total	0	1	2	3	4	5	≥ 6	Total
Eample 10	DI	**100**	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100
	PDI	**100**	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100	100	0	0	0	0	0	0	100
	E-div	**0**	0	0	1	0	0	8	9	1	1	0	0	0	1	9	12	0	0	0	1	0	0	2	3
	Kernel	**2**	7	7	6	4	3	71	100	1	5	2	6	1	3	82	100	3	6	3	2	2	8	75	99
	MST	**1**	2	2	3	3	0	12	23	2	2	2	1	1	0	22	30	2	4	1	3	0	1	21	32
	NNG	**2**	1	4	4	2	0	8	21	5	5	2	0	2	1	19	34	2	3	2	2	0	3	8	20
	MDP	**4**	3	0	2	1	3	26	39	3	2	1	0	1	0	25	32	5	1	1	1	2	4	21	35
Eample 11	DI	**35**	0	0	0	0	0	45	80	48	0	0	0	0	4	37	89	34	0	0	0	0	0	48	82
	PDI	**30**	0	0	0	0	12	40	82	43	0	5	6	5	0	27	87	30	0	3	4	2	5	46	90
	E-div	**0**	0	0	0	0	0	6	6	2	0	0	1	0	0	2	5	0	1	0	0	0	0	3	4
	Kernel	**0**	5	8	4	2	4	76	99	2	3	3	3	2	5	82	100	3	3	5	6	7	2	71	97
	MST	**0**	0	0	0	0	0	9	9	0	0	0	0	0	0	20	20	0	0	1	0	1	1	17	20
	NNG	**0**	0	0	0	0	0	12	12	0	0	0	0	0	0	9	9	1	1	1	0	1	0	11	15
	MDP	**0**	0	0	0	0	0	12	12	0	0	0	0	0	0	18	18	0	0	0	0	0	0	13	13

Table: Frequency distribution of $|\hat{\tau}-\tau|$,where $\hat{\tau}$is the predicted location of change point

- **Example 12**: $X_1, X_2, \cdots, X_{15} \sim N_{250}(0.5\mathbf{1}_{250}, \mathbf{I_{250}})$,
  $X_{16}, X_{17}, \cdots, X_{30} \sim N_{250}(0_{250}, \mathbf{I_{250}})$,
  $X_{31}, X_{42}, \cdots, X_{45} \sim N_{250}(0.5\mathbf{1}_{250}, \mathbf{I_{250}})$,
- **Example 13**: $X_1, X_2, \cdots, X_60$ observations from four Gaussian distributions $N_{250}(0_{250}, (\frac{1}{20})^{i-1}\mathbf{I_{250}})$, i = 1,2,3,4,differing only their scales. The first 15 observations are generated from first distribution $N_{250}(0_{250}, \mathbf{I_{250}})$,next 15 from the second and so on .

Result of Example 12



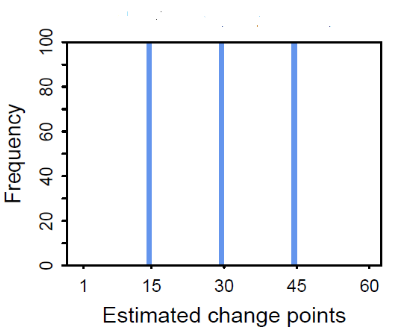
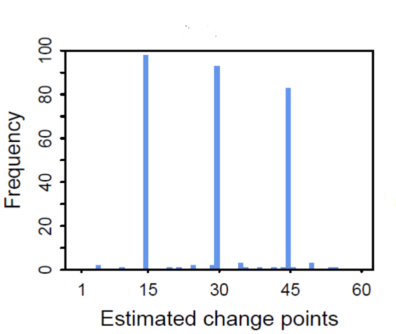


Figure: E-divisive and Kernel method



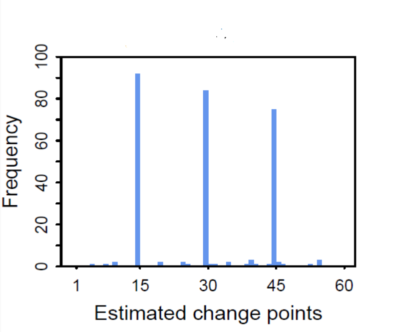


Figure: DUNN index and Penalized DUNN index

Result of Example 13

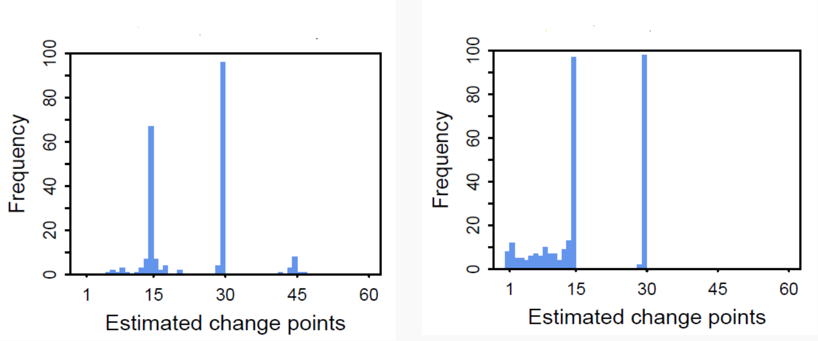
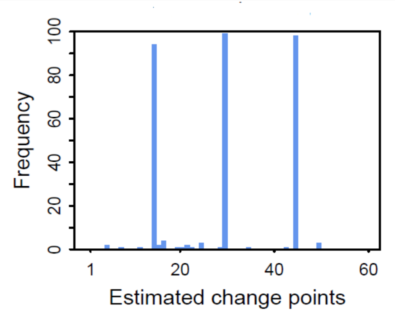


Figure: E-divisive and Kernel method



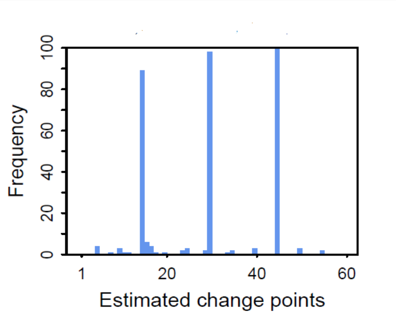


Figure: DUNN index and Penalized DUNN index

Results from the analysis of Real data

- We apply our proposed method to the Reality Mining data set gathered during a research study conducted by the MIT Media Laboratory. It contains information on the caller, the callee and the time for every call made during the period from July 20, 2004, to June 14, 2005, i.e. 48 weeks. The data set is available on the website(`http://realitycommons.media.mit.edu/realitymining.html`).

Comparing the result with the existing algorithm

- DI and PDI both identified week 22 (December 14-20, 2004) as the change point.
- E-divisive method recognized week 7 (August 31 - September 06, 2004) as the change point
- The kernel method detected week 36 (March 22-28, 2005) as the pivotal period.
- The graph-based techniques, including MST, NNG, and MDP, the change-point was detected at week 41 (April 26 - May 02, 2005), week 33 (February 28 - March 07, 2005), and week 4 (August 10-16, 2004), correspondingly.

Note: But, week 22 was the beginning of the winter break when the students and staff were moving from the fall session to the spring session. This provides a strong justification for the selection of this week as a change point in the friendship network pattern.

Conclusion

- Another clustering algorithm (e.g. K-means, spectral clustering based on different types of linkages) can also be used.
- It is not yet known how these methods will perform if the sample size grows with the dimension.

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Thank you

Proof of Theorem1

Proof

Case(1): *For any* $k < K_{opt}$*, there is at least one cluster that includes observations from two distinct populations and no two clusters contain observations from the same population.*
Under the assumption $(A1)$ *we get,* $W_k(i) \xrightarrow{p} \psi_+(d)$*,for some i.Hence* $W(k) = max_{1 \leq i \leq k} W_i(k) \xrightarrow{p} \psi_+(d)$*.Also,* $B_k(i,j) \xrightarrow{p} \psi_+(d)$ *for every* $i \neq j$*.Hence,* $B(k) = min_{1 \leq i < j \leq k} B_k(i,j) \xrightarrow{p} \psi_+(d)$*.So,*
$D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} C_1$*(some constant).*
Case(2): *For any* $k > K_{opt}$*,no cluster contains observations from different populations and at least two clusters contain observations from the same populations.*
Under the assumptions we get, $B_k(i,j) \xrightarrow{p} \psi_-(d)$*,for some* $i \neq j$*.Hence* $B(k) = min_{1 \leq i < j \leq k} B_k(i,j) \xrightarrow{p} \psi_-(d)$ *and* $W_k(i) \xrightarrow{p} \psi_-(d)$ *for every* i*.Hence,* $W(k) = max_{1 \leq i \leq k} W_k(i) \xrightarrow{p} \psi_-(d)$*.So,*
$D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} C_2$*(some constant).*

Proof

Case(3): *For any $k = K_{opt}$, each cluster contains observations from same population. Two different clusters contain observations from two different populations.*

So,under A1,we get,$W_k(i) \xrightarrow{p} \psi_{-}(d)$,for every i.Hence

$W(k) = max_{1 \leq i \leq k} W_i(k) \xrightarrow{p} \psi_{-}(d)$.Also $B_k(i,j) \xrightarrow{p} \psi_{+}(d)$ for every $i \neq j$.Hence,$B(k) = min_{1 \leq i < j \leq k} B_k(i,j) \xrightarrow{p} \psi_{+}(d)$.So,

$D(k) = \frac{B(k)}{W(k)} \xrightarrow{p} \infty$.

Now, $Pr(D(k_{opt}) > D(k) \forall k \neq k_{opt}) \xrightarrow{p} 1 \implies \hat{K_D} \xrightarrow{p} k_{opt}$ as $d \to \infty$

Proof of Theorem2

Proof

Case(1): *For any* $k \geq 2$,

$$D(k) \xrightarrow{p} \begin{cases} \frac{\psi_+(d)}{\psi_-(d)} & \text{for } k = k_{opt} \\ C & \text{for } k \neq k_{opt} \end{cases}$$

where C is constant

For $k \neq k_{opt}$, $PD(k_{opt}) - PD(k) = D(k_{opt}) - k_{opt}\zeta(d) - D(k) + k\zeta(d) = D(k_{opt}) - D(k) - (k_{opt} - k)\zeta(d) \xrightarrow{p} (\frac{\psi_+(d)}{\psi_-(d)} - C) - (k_{opt} - k)\zeta(d) \xrightarrow{p} \infty$*(Since,*$\zeta(d) =$ ***o***$(\frac{\psi_+(d)}{\psi_-(d)})$ *as* $d \to \infty$*).Thus,* $\hat{K_{PD}} \xleftarrow{p} k_{opt}$

Case(2): *When* $k_{opt} = 1$*,we have* $W(k) \xrightarrow{p} \psi_-(d), B(k) \xrightarrow{p} \psi_-(d)$ *for all* $k \geq 1$*.Therefore,* $D(k) \xrightarrow{p} C$*.For* $k \neq 1$, $PD(1) - PD(k) = D(1) - \zeta(d) - D(k) + k\zeta(d) = D(1) - D(k) + (k-1)\zeta(d) \xrightarrow{p} \infty$*, as* $d \to \infty$*. Hence* $\hat{K_{PD}} \xrightarrow{p} \infty$ *as* $d \to \infty$

Contd..

Using the result of the lemma,
Let X_1, X_2, Y_1, Y_2 be independent real random variables with finite expectations. Let X_1, X_2 be identically distributed with distribution function F and let Y_1, Y_2 be identically distributed with distribution function G. Then

$$2E\left\|X_1 - Y_1\right\| - E\left\|X_1 - X_2\right\| + E\left\|Y_1 - Y_2\right\| \geq 0$$

where the equality holds if and only if $F = G$ Here, we use a one-dimensional version of the above expression
Let $X_1^{(i)}, X_2^{(i)} \sim F^{(i)}$ and $Y_1^{(i)}, Y_2^{(i)} \sim G^{(i)}$

$$2E|X_1^{(i)} - Y_1^{(i)}| - E|X_1^{(i)} - X_1^{(i)}| - E|Y_1^{(i)} - Y_2^{(i)}| \geq 0$$

$$d^{-1}(n-2)\delta_1(X_i, X_j) \xrightarrow{\text{p}} \begin{cases} 0 & \text{for } 1 \leq i, j \leq \tau \\ 0 & \text{for } \tau < i, j \leq n \\ d_{12}^{\star\star} & \text{for } i \leq \tau < j \text{ or } j \leq \tau < i, \end{cases}$$

where $d_{12}^{\star\star} \geq 0$ and the equality holds if and only if F and G have identical marginal distributions.